

Energy Methods for the Wave Equation

Conservation, Uniqueness, and Finite Propagation

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The 1D Wave Equation

$$u_{tt} - c^2 u_{xx} = 0, \quad x \in \mathbb{R}, \quad t > 0$$

Initial data:

$$u(x, 0) = \phi(x), \quad u_t(x, 0) = \psi(x)$$

- Hyperbolic PDE
- Finite propagation speed
- Energy structure

Defining the Energy

Multiply the equation by u_t :

$$u_{tt}u_t - c^2 u_{xx}u_t = 0.$$

Integrate over $\mathbb{R}$:

$$\int_{\mathbb{R}} u_{tt}u_t \, dx - c^2 \int_{\mathbb{R}} u_{xx}u_t \, dx = 0.$$

Computing the Terms

First term:

$$\int u_{tt} u_t \, dx = \frac{1}{2} \frac{d}{dt} \int u_t^2 \, dx.$$

Second term (integration by parts):

$$\int u_{xx} u_t \, dx = - \int u_x u_{xt} \, dx.$$

Therefore:

$$\int u_{xx} u_t \, dx = - \frac{1}{2} \frac{d}{dt} \int u_x^2 \, dx.$$

Energy Identity

Putting everything together:

$$\frac{d}{dt} \left[\frac{1}{2} \int (u_t^2 + c^2 u_x^2) dx \right] = 0.$$

Energy

$$E(t) = \frac{1}{2} \int_{\mathbb{R}} (u_t^2 + c^2 u_x^2) dx.$$

$$E(t) = E(0)$$

Energy is conserved.

$$E(t) = \frac{1}{2} \int u_t^2 + \frac{c^2}{2} \int u_x^2$$

- u_t^2 = kinetic energy
- $c^2 u_x^2$ = potential energy
- Energy transfers between the two
- Total energy remains constant

Uniqueness via Energy

Suppose u solves the homogeneous problem with

$$u(x, 0) = 0, \quad u_t(x, 0) = 0.$$

Then

$$E(0) = 0.$$

By conservation:

$$E(t) = 0.$$

Thus

$$u_t = 0, \quad u_x = 0.$$

Therefore $u \equiv 0$.

Localized Energy

Define energy in an interval:

$$E_R(t) = \frac{1}{2} \int_{|x| \leq R} (u_t^2 + c^2 u_x^2) dx.$$

Energy flux across boundaries determines how influence spreads.

Differentiate:

$$\frac{d}{dt}E_R(t) = -c^2 u_x u_t \Big|_{x=-R}^{x=R}.$$

- Energy changes only via boundary flux
- If initial data vanish in $|x| \leq R - ct$
- Then solution remains zero in $|x| \leq R - ct$

Finite Speed of Propagation

Conclusion

Signals travel at speed at most c .

Value at (x, t) depends only on data in

$$[x - ct, x + ct].$$

This matches the characteristic picture.

Forced Wave Equation

$$u_{tt} - c^2 u_{xx} = f(x, t).$$

Multiply by u_t and integrate:

$$\frac{d}{dt} E(t) = \int f u_t \, dx.$$

Energy Inequality

Using Cauchy–Schwarz:

$$\left| \int f u_t dx \right| \leq \|f\|_{L^2} \|u_t\|_{L^2}.$$

Thus:

$$\frac{d}{dt} E(t) \leq C \|f\|_{L^2}^2 + C E(t).$$

Gronwall's inequality yields bounds on growth.

Dirichlet Boundary Conditions

On $(0, L)$ with

$$u(0, t) = u(L, t) = 0,$$

integration by parts gives:

$$E'(t) = 0.$$

Energy conserved under fixed boundary conditions.

Neumann Boundary Conditions

If

$$u_x(0, t) = u_x(L, t) = 0,$$

boundary terms vanish again.

Energy is conserved.

Final Summary

- Multiply by u_t to obtain energy identity
- Energy is conserved for homogeneous wave equation
- Energy implies uniqueness
- Energy flux explains finite propagation
- Forcing introduces energy growth estimates